

Morning Paper Report - Tuesday, June 30, 2026

Recent, non-repeated papers matched to visual working memory, visual search, modeling, working-memory representation, neural-network memory models, AI for human memory, and egocentric predictive-coding memory themes.

1. Investigating the effects of context, visual working memory, and inhibitory control in hybrid visual search.

Barbosa A, Ruarte G, Ries AJ, Kamienkowski JE, Ison MJ
Frontiers in human neuroscience, 2024

Source: PubMed; matched terms: visual working memory, visual search, working memory

Link: <https://pubmed.ncbi.nlm.nih.gov/39257697/>

Goal: The title and metadata suggest relevance to Lila's working-memory and attention topics, but no abstract was available from the source API.

Method: Method details were not available in the retrieved metadata.

Results: Result details were not available in the retrieved metadata.

Conclusion: Open the linked record to inspect the full abstract or paper before journal-club use.

2. The locus of visual search's interference on visual working memory.

Hamblin-Frohman Z
Journal of experimental psychology. Human perception and performance, 2022

Source: PubMed; matched terms: visual working memory, visual search, working memory

Link: <https://pubmed.ncbi.nlm.nih.gov/35925748/>

Goal: The title and metadata suggest relevance to Lila's working-memory and attention topics, but no abstract was available from the source API.

Method: Method details were not available in the retrieved metadata.

Results: Result details were not available in the retrieved metadata.

Conclusion: Open the linked record to inspect the full abstract or paper before journal-club use.

3. Measurement Models for Visual Working Memory – A Factorial Model Comparison

Klaus Oberauer
2021

Source: Crossref; matched terms: visual working memory, working memory, precision, modelling

Link: <https://doi.org/10.31234/osf.io/9vwhx>

Goal: Several measurement models have been proposed for data from the continuous-reproduction paradigm for studying visual working memory: The original mixture model (Zhang & Luck, 2008) and its extension (Bays, Catalao, & Husain, 2009); the interference measurement model (Oberauer, Stoneking, Wabersich, & Lin, 2017), and the target confusability competition model (Schurgin, Wixted, & Brady, 2020)

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target confusability competition model (Schurgin, Wixted, & Brady, 2020)

Results: Fitting all models to eight data sets revealed a new model as empirically most adequate, which combines a von-Mises activation function with a signal-detection response-selection rule

Conclusion: The precision parameter can be treated as a constant across many experimental manipulations, though it might vary with manipulations not yet explored All modelling code and the raw data modelled are available on the OSF: osf.io/zwprv

4. Computational Models Based on Working Memory

Kayra Ügümü

Human Computer Interaction, 2024

Source: Crossref; matched terms: working memory, computational model, computational models

Link: <https://doi.org/10.62802/t6qvkr65>

Goal: Understanding the intricacies of working memory (WM) is crucial for unraveling cognitive processes

Method: This paper explores the relationship between working memory and computational neuroscience, aiming to shed light on how computational models contribute to our comprehension of WM

Results: This paper explores the relationship between working memory and computational neuroscience, aiming to shed light on how computational models contribute to our comprehension of WM The discussion delves into the processes involved in working memory, the computational principles that underlie it, and the potential downsides of relying on computational models.

Conclusion: This paper explores the relationship between working memory and computational neuroscience, aiming to shed light on how computational models contribute to our comprehension of WM The discussion delves into the processes involved in working memory, the computational principles that underlie it, and the potential downsides of relying on computational models.

5. A certain future strengthens the past: Knowing ahead how to act on an object prioritizes its Visual Working Memory representation

Authors unavailable

2023

Source: OSF/PsyArXiv; matched terms: visual working memory, working memory representation, working memory

Link: https://osf.io/preprints/psyarxiv/g6894_v1/

Goal: Findings from recent studies indicate that planning an action towards an object strengthens its visual working memory (VWM) representation, emphasizing the importance of sensorimotor links in VWM

Method: In three experiments, we asked participants to memorize a visual stimulus for a subsequent memory test, whereby they performed a specific hand movement towards memory-matching probes

Results: Across the three experiments, we found moderate evidence that knowing in advance how to act on an object prioritized its mnemonic representation, as expressed in an increased attentional bias towards it

Conclusion: Findings from recent studies indicate that planning an action towards an object strengthens its visual working memory (VWM) representation, emphasizing the importance of sensorimotor links in VWM